Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			Is the [resultant] force for circular motion (1) It is directed towards the centre [of the circle] (1)	2			2		
	(b)	(i)		$T = \frac{15}{10} = 1.5$ s **or** $f = \frac{10}{15} = 0.67$ [Hz] (1) $v = \frac{2\pi r}{T} = \frac{2\pi(0.8)}{1.5}$ (1)[= 3.35 m s^{-1}] **Alternative:** Total distance = $1.6\pi \times 10$ (1) $v = \frac{1.6\pi \text{x} 10}{15}$ (1) [= 3.35 m s^{-1}]	1	1		2	2	
		(ii)		Substitution: $F = \frac{mv^2}{r} = \frac{(30 \times 10^{-3})\text{x}\, 3.35^2}{0.8}$ (1) $= 0.42$ [N] (1) **Alternative:** $F = mr\omega^2 = (30 \times 10^{-3}) \times 0.8 \times \left(\frac{2\pi}{1.5}\right)^2$ (1) $= 0.42$ [N](1)	1	1		2	2	
	(c)			Force = Tension + mg (1) Tension = Force $-$ mg = 0.42 **ecf** $-$ $(30 \times 10^{-3})(9.81) = 0.13$ [N] (1) Award 1 mark for 0.7 [N]		2		2	1	